



Case Study II: OptiKuh Dairy Cow Health Data

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What we did

- ▶ Performed exploratory data analysis on:
 - ▶ dataset size and observation period
 - ▶ missing values and measurement availability
 - ▶ health-status distribution
 - ▶ farm-level and breed-level differences
 - ▶ diagnosis labels
 - ▶ milk yield and metabolic-marker patterns

Dataset Overview

- ▶ Daily records: 492,540
- ▶ Original variables: 49
- ▶ Unique animals: 1,714
- ▶ Anonymized farms: 12
- ▶ Approx. lactation episodes: 3,422
- ▶ Observation window: 2014-08-28 to 2017-03-23

Note: this is a large repeated-measurement dataset, not a simple one-row-per-animal dataset.

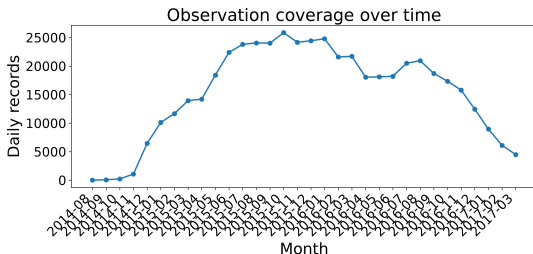
Basic Data Quality Checks

- ▶ Dates were parsed to check the 32-month observation window
- ▶ Animal, farm and lactation identifiers were checked for repeated measurements
- ▶ Missing values were checked column-wise and by measurement group
- ▶ Health status was summarized at lactation-episode level to avoid repeated counting
- ▶ Daily-record summaries and lactation-episode summaries were kept separate

Main point: the data must be handled according to animal, farm and lactation structure.

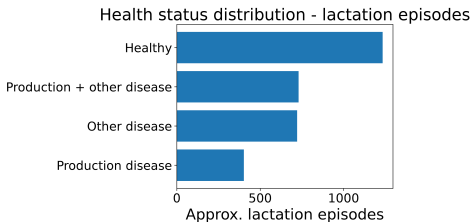
Analysis 1.1

- Observation period covers 32 months
- Monthly records range from 4 rows at the beginning to 25,852 rows at the peak
- Highest monthly coverage was in 2015-10



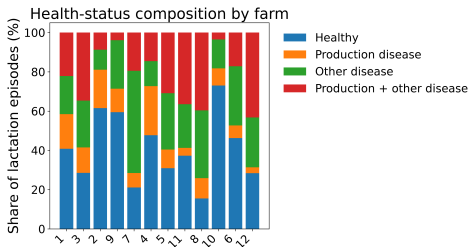
Analysis 1.2

- Healthy: 1,547 episodes, 45.21%
- Production disease only: 404 episodes, 11.81%
- Other disease only: 740 episodes, 21.62%
- Production disease + other disease: 731 episodes, 21.36%



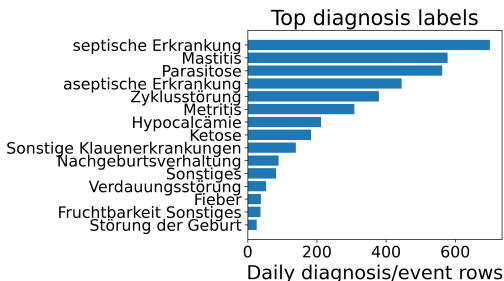
Analysis 1.3

- ▶ Health-status composition differs across all 12 farms
- ▶ Farm episode counts range from 84 to 722 episodes
- ▶ The largest farm has about 8.6 times more episodes than the smallest farm



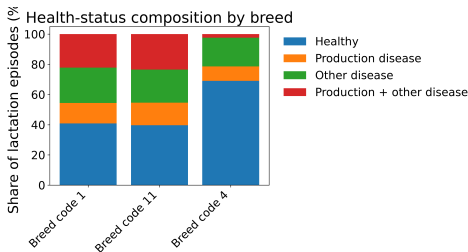
Analysis 1.4

- ▶ Top diagnosis: septic claw/hoof disease with 700 event rows
- ▶ Mastitis: 578 event rows
- ▶ Parasitic disease: 562 event rows
- ▶ Top 4 diagnoses together account for 2,285 event rows



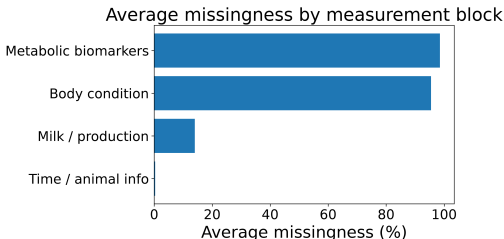
Analysis 1.5

- ▶ Breed groups show different health-status distributions
- ▶ German Holstein is the dominant breed group
- ▶ Smaller breed groups have fewer lactation episodes and should be interpreted carefully
- ▶ Breed should be retained as a grouping variable in future analysis



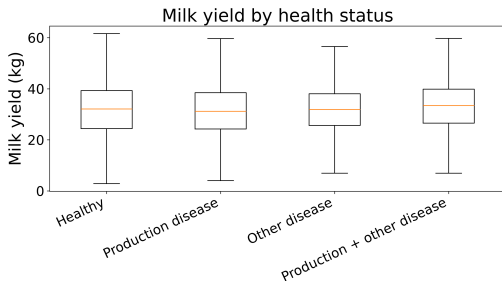
Analysis 1.6

- ▶ Milk yield is missing in 13.98% of daily records
- ▶ Body condition score is missing in 96.22% of daily records
- ▶ BHB, NEFA and glucose are missing in about 98.53% of daily records
- ▶ Missingness mainly reflects different measurement schedules



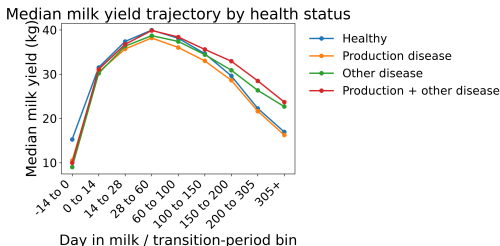
Analysis 1.7

- Milk yield is available in 423,673 daily records
- Overall median milk yield is 32.45 kg
- Median milk yield differs by health status
- Lowest and highest health-status medians differ by about 2.14 kg



Analysis 1.8

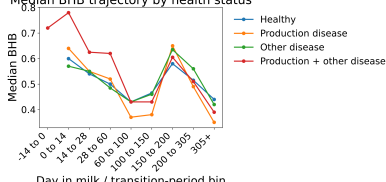
- Milk yield changes across days-in-milk bins
- The time trend differs between health-status groups
- Days in milk should be considered in feature engineering
- Raw milk yield should not be interpreted without lactation-stage context



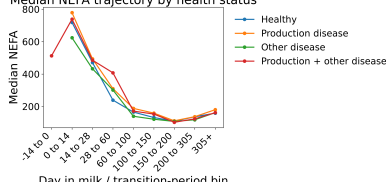
Analysis 1.9

- ▶ BHB above about 1.2 mmol/L is commonly used as a subclinical ketosis warning threshold
- ▶ NEFA above about 0.7 mmol/L after calving can indicate excessive fat mobilization
- ▶ Both markers are linked to negative energy balance in early lactation
- ▶ Interpretation depends strongly on sampling time around calving

Median BHB trajectory by health status



Median NEFA trajectory by health status



Agenda for next week

- ▶ Define the final analysis unit:
 - ▶ daily record vs. lactation episode
- ▶ Define the production-disease target clearly
- ▶ Decide the time window around calving
- ▶ Build preprocessing steps for sparse variables
- ▶ Prepare the first baseline modelling dataset